

1. A bin of 50 parts contains 5 that are defective. A sample of 10 parts is selected without replacement. How many samples contain at least 4 defective parts?

2. The probability is 1% that an electrical connector that is kept dry fails during the warranty period. If the connector is ever wet, the probability of a failure during the warranty period is 5%. If 90% of the connectors are kept dry and 10% are wet, what proportion of connectors fail during the warranty period?

3. A computer system uses passwords that contain exactly eight characters, and each character is one of 26 lowercase letters (a–z) or 26 uppercase letters (A–Z) or 10 integers (0–9). Let Ω denote the set of all possible passwords, and let A and B denote the events that consist of passwords with only letters or only integers, respectively. Suppose that all passwords in Ω are equally likely. Determine the probability of each of the following:

- a. A
- b. B
- c. A password contains at least 1 integer.
- d. A password contains exactly 2 integers.

4. Because all airline passengers do not show up for their reserved seat, an airline sells 125 tickets for a flight that holds only 120 passengers. The probability that a passenger does not show up is 0.10, and the passengers behave independently. What is the probability that every passenger who shows up can take the flight?

5. A player of a video game is confronted with a series of opponents and has an 80% probability of defeating each one. Success with any opponent is independent of previous encounters. Until defeated, the player continues to contest opponents.

- a. What is the probability mass function of the number of opponents contested in a game?
- b. What is the probability that a player defeats at least two opponents in a game?

6. A state runs a lottery in which six numbers are randomly selected from 40 without replacement. A player chooses six numbers before the state's sample is selected.

- a. What is the probability that the six numbers chosen by a player match all six numbers in the state's sample?
- b. What is the probability that five of the six numbers chosen by a player appear in the state's sample?

7. The diameter of a particle of contamination (in micrometers) is modeled with the probability density function $f(x) = 2x^{-3}$ for $x > 1$. Determine the following:

- a. $P(X < 2)$
- b. $P(X > 5)$
- c. $P(4 < X < 8)$
- d. $P(X < 4 \text{ or } X > 8)$
- e. x such that $P(X < x) = 0.95$

8. The thickness of a flange on an aircraft component is uniformly distributed between 0.95 and 1.05 millimeters. Determine the following:

- a. Proportion of flanges that exceeds 1.02 millimeters
- b. Thickness exceeded by 90% of the flanges

1. Exactly 4 defective parts + exactly 5 defective parts = $C(5, 4) \times C(45, 6) + C(5, 5) \times C(45, 5) = 40725300 + 1221759 = 41947059$

2. Let F denote the event that a connector fails. Let W denote the event that a connector is Wet. $\rightarrow W'$ is a connector that kept dry. $P(F) = P(F|W)P(W) + P(F|W')P(W') = 0.05 \times 0.1 + 0.01 \times 0.9 = 0.014$

3. (a) $P(A) = (52^8)/(62^8) = 0.2448461$

(b) $P(B) = (10^8)/(62^8) = 4.580011 \times 10^{-7}$

(c) $P(\text{at least 1 integer}) = 1 - P(\text{no integer}) = 1 - P(A) = 1 - 0.2448461 = 0.7551539$

(d) $P(\text{exactly 2 integers}) = (C(8, 2) \times (10^2) \times (52^6))/(62^8) = 0.2535389$

4. $P(X \geq 5) = 1 - P(X < 5) = 1 - F(4) = 1 - 0.00386 = 0.99614$

$F(4) = P(X \leq 4) = \text{*insert binomial math here*} = 0.00386$

5. (a) $X \sim \text{Geo}(0.2) \rightarrow f(x) = 0.8^{x-1} \times 0.2, x = 1, 2, 3, \dots$

(b) $P(X > 2) = 1 - F(2) = 1 - P(X = 1) - P(X = 2) = 1 - 0.2 - 0.8 \times 0.2 = 0.64$

6. (a) $P(X = 6) = f(6) = (C(6, 6)C(34, 0))/(C(40, 6)) = 1/3838380 = 2.605266 \times 10^{-7}$

(b) $P(X = 5) = f(5) = (C(6, 5)C(34, 1))/(C(40, 6)) = (6 \times 34)/3838380 = 5.31474 \times 10^{-5}$

7. (a) $P(X < 2) = F(2) = 1 - 1/4 = 3/4$

(b) $P(X > 5) = 1 - F(5) = 1/25$

(c) $P(4 < X < 8) = F(8) - F(4) = 1/16 - 1/64 = 3/64 = 0.046875$

(d) $P(X < 4 \text{ or } X > 8) = 1 - P(4 \leq X \leq 8) = 1 - 0.046875 = 0.953125$

(e) $F(x) = 1 - 1/x^2 = 0.95 \rightarrow x^2 = 1/0.05 = 20 \rightarrow x = \sqrt{20} = 4.472136$

8. Let X denote the thickness of a flange on an aircraft component. $\rightarrow X \sim \text{Unif}(0.95, 1.05) \rightarrow f(x) = 10$

(a) $P(X > 1.02) = 1 - F(1.02) = 1 - 0.7 = 0.3$

(b) $P(X > x) = 0.9 \rightarrow 1 - F(x) = 0.9 \rightarrow F(x) = 0.1 \rightarrow 10x - 9.5 = 0.1 \rightarrow x = 0.96$